

SignDino: Self-Supervised Sign Language Representation Learning via Temporal-Axis Self-Distillation

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Abstract

Self-supervised sign language representation learning must model two properties not central to natural-image SSL: signs are produced by a small set of anatomically distinct articulators, and their meaning depends on the temporal organisation of those articulators. We introduce SIGNDINO, a self-supervised sign-video encoder that moves the DINOv3 student-teacher recipe from the spatial domain of image crops to the temporal domain of tracked sign streams. Each video is decomposed into left-hand, right-hand, and face streams by a detector-first YOLOv8n+ByteTrack pipeline. A frozen DINOv3 ViT-B/16 embeds each per-frame anatomical crop, while lightweight temporal Transformers, not the image backbone, form the student and EMA teacher. They are trained by temporal DINO self-distillation, frame-level masked-token prediction in the style of iBOT, KoLeo feature spreading, and Gram anchoring of the frame-to-frame similarity structure. This design keeps strong image-level visual primitives fixed and learns only how articulator states evolve across time. We evaluate on sign-to-English translation, isolated sign recognition, and fingerspelling detection benchmarks. Across these tasks, SIGNDINO provides a strong public self-supervised representation and shows competitive or state-of-the-art performance under matched downstream evaluation.

Project page:

<https://junyi2005.github.io/signdino/>

Code:

<https://github.com/junyi2005/signdino>

1 Introduction

Sign language is a temporally organised visual language produced through coordinated articulators: the hands move asymmetrically, the face carries grammatical and affective information, and short

holds or repetitions can change lexical meaning. A sign-video representation learner must therefore capture both *where* linguistic evidence is articulated and *how* it evolves over time. Whole-frame video SSL wastes much of its capacity on background and signer appearance, while pose-only SSL discards hand-shape, contact, and mouth cues and inherits keypoint-extractor failures on fast motion (§3.1). We instead represent each frame by three tracked anatomical cues,

$$\text{cue}_t = \{x_t^L, x_t^R, x_t^F\}, \quad t = 1, \dots, T, \quad (1)$$

where L , R , and F denote the left hand, right hand, and face. The central question is how to learn from these streams without gloss labels, translations, or hand-defined sub-sign units.

The DINO family offers an incomplete analogy. Image DINO (Caron et al., 2021) trains a student to match a momentum teacher under different spatial crops of an image; DINOv2 (Oquab et al., 2024) and DINOv3 (Siméoni et al., 2025) scale and stabilise this recipe for dense visual representation learning. In sign video, the natural counterpart of a local image crop is not another spatial view of the same frame but a shorter temporal view of the same tracked articulator. A hand crop at frame t is meaningful because it belongs to a trajectory: its identity depends on the preceding configuration, the following motion, and its coordination with the other hand and the face. We therefore transpose DINO’s local–global consistency from image space to the time axis (Figure 1): the teacher observes a longer temporal crop of one stream, the student shorter, partially masked crops, and learns to match the teacher’s prototype and frame-similarity structure.

Recent sign-specific SSL makes this concrete. SHUBERT (Gueuwou et al., 2025b), an SSL baseline for American Sign Language, predicts offline-clustered hidden units for face, hand, and body-pose streams; its supervision comes from precomputed clusters rather than local–global temporal

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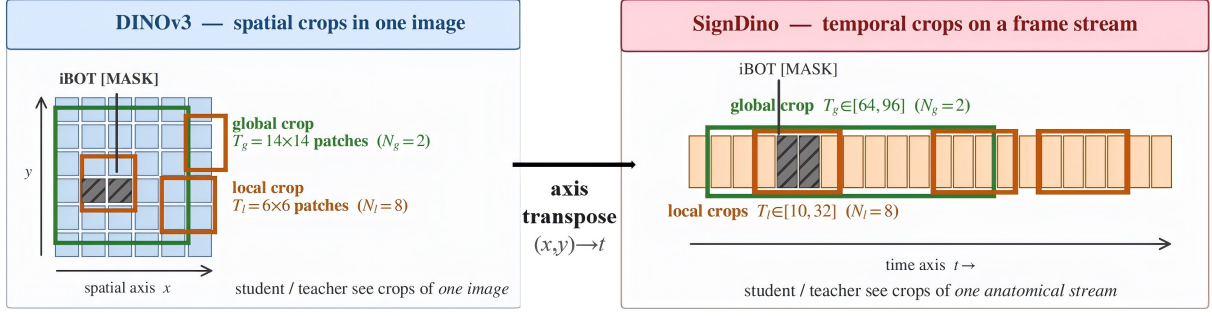


Figure 1: The local–global distillation game, ported from image space to the time axis. *Left*: image DINOv3 (Siméoni et al., 2025) samples spatial global and local crops in a *single image*. *Right*: SIGNDINO samples temporal global ($T_g \sim \mathcal{U}\{64, \dots, 96\}$) and local ($T_l \sim \mathcal{U}\{10, \dots, 32\}$) crops along the tracked L_t, R_t, F_t rows of a *single anatomical stream*, with iBOT masks on a few frame slots inside the local crop.

agreement. Other sign SSL relies on hand-pose reconstruction (Hu et al., 2021, 2023), masked-video reconstruction (Rust et al., 2024; Zhao et al., 2024), or sign–text contrastive alignment (Jiang et al., 2024). SIGNDINO (Figure 2) keeps the multi-stream factorisation of SHUBERT but changes the pretext task: each stream is trained by continuous student–teacher distillation over temporal crops, with iBOT on masked frame tokens and Gram anchoring preserving frame-to-frame structure. We evaluate under the SHUBERT protocol on sign-to-English translation (How2Sign (Duarte et al., 2021), OpenASL (Shi et al., 2022), FLEURS-ASL (Tanzer, 2024b)), isolated sign recognition (ASL Citizen (Desai et al., 2023), Sem-Lex (Kezar et al., 2023), WLASL2000 (Li et al., 2020)), and fingerspelling detection on ASL-STEM Wiki (Yin et al., 2024).

Our contributions are: (i) We formulate *temporal-axis DINO* for sign-language video: per-frame articulator tokens replace spatial patch tokens, multi-temporal-crop replaces multi-spatial-crop, iBOT masks frames rather than patches, and Gram anchoring constrains a clip’s frame-to-frame structure. (ii) We define a three-stream anatomical SSL design in which left hand, right hand, and face are encoded independently over time and fused only downstream, so each stream learns its own dynamics while fusion stays transparent. (iii) We introduce a detector-first anatomical crop pipeline using YOLOv8n and ByteTrack, reducing reliance on pose-driven crops for fast or occluded hand motion (§3.1).

2 Related Work

SSL across modalities and the DINO family. Self-supervised pre-training has progressed from

masked-token text (Devlin et al., 2019; Liu et al., 2019; Lan et al., 2020; Mohamed et al., 2022) to continuous-signal speech (Baevski et al., 2020; Hsu et al., 2021; Yang et al., 2021) to image SSL (Caron et al., 2020; Radford et al., 2021; Caron et al., 2021) centred on the DINO student-teacher paradigm. DINOv2 (Oquab et al., 2024) and DINOv3 (Siméoni et al., 2025) scale it with Sinkhorn–Knopp normalisation (Caron et al., 2020), KoLeo regularisation (Sablayrolles et al., 2019) and Gram anchoring; iBOT (Zhou et al., 2022; He et al., 2022) adds masked-patch prediction, and VideoMAE (Tong et al., 2022) ports MAE to video. We define each of these components where we use it, in §3.2. SIGNDINO freezes a DINOv3 encoder and re-applies its DINO+iBOT+DKoleo+Gram recipe along each tracked stream’s time axis.

Sign-language representation learning. Supervised SLT pre-trains on parallel video–text and gloss corpora (Camgoz et al., 2018, 2020; Graves et al., 2006; Zhou et al., 2021; Zuo et al., 2023, 2024; Uthus et al., 2023; Tanzer and Zhang, 2024; Zhang et al., 2024) and recently pairs visual encoders with LLM decoders for gloss-free translation (Li et al., 2025; Gueuwou et al., 2025a; Hwang et al., 2025; Chen et al., 2025; Lin et al., 2023; Wong et al., 2024). Prior sign SSL targets single articulators or whole frames (Hu et al., 2021, 2023; Zhao et al., 2024; Rust et al., 2024; Jiang et al., 2024; Jiao et al., 2024). Our closest peer SHUBERT (Gueuwou et al., 2025b) ports HuBERT to four sign streams with masked cluster prediction on 984 hours of YouTube-ASL; SIGNDINO keeps the factorisation but swaps discrete clusters for continuous student/teacher distillation along the time axis.

Multi-stream pipelines, detector-first tracking, fingerspelling. Multi-stream sign architectures date back to TwoStream-SLR and STMC (Chen et al., 2022; Yin and Read, 2020; Cheng et al., 2023). Single-frame detectors fail under motion blur and self-occlusion, so ByteTrack (Zhang et al., 2022) Kalman-fills dropped frames. Fingerspelling is a dense-temporal-cue sub-task (Hanson, 1982; Shi et al., 2019; Fayyazsanavi et al., 2024; Tanzer, 2024a; Yin et al., 2024), and phonological decomposition of signs is operationalised by ASL-Lex 2.0 and ASL Citizen (Sehyr et al., 2021; Desai et al., 2023, 2024).

3 SignDino

SIGNDINO comprises three independently trained per-stream SSL encoders sharing one upstream anatomical-crop pipeline (Figure 2). Throughout, an *anatomical crop* is the image region of one articulator (left hand, right hand, or face) in one frame, not a ViT spatial patch; a *temporal crop* is a contiguous block of frames of one such stream, and is the object the student–teacher game is played on. The frozen DINOv3 encoder in §3.1 embeds each crop into a per-frame vector; student and teacher (Figure 4) are temporal Transformers over these embeddings. Streams are encoded independently during SSL, postponing fusion to the downstream head.

3.1 Multi-Stream Feature Pre-Processing

Stage P1: detector-first crop extraction (YOLOv8n + ByteTrack). A YOLOv8n hand+face detector emits per-frame boxes, and a ByteTrack (Zhang et al., 2022) tracker associates hand identities across frames, recovering dropouts of $\leq K$ frames by track-state interpolation. Face crops use $1.2\times$ padding and hand crops $1.4\times$ (larger context absorbs motion blur). Crops are written at 112×112 and resized to 224^2 at SSL time. When tracking fails for more than K consecutive frames, the matching entry of a per-frame validity mask $m^{(r)} \in \{0, 1\}^T$ is set to 0 and the SSL loss masked there. The Kalman-filled tracker reaches $\sim 95\%$ per-frame hand detection on How2Sign; Figure 3 shows the tracker recovering hands YOLOv8n alone misses, and §4.6 quantifies the downstream impact.

Stage P2: frozen DINOv3 per-frame embedding. Each crop is forwarded through a frozen DINOv3

ViT-B/16 (Siméoni et al., 2025), taking the CLS output as the 768-d per-frame representation:

$$e_t^{(r)} = \text{DINOv3-CLS}(x_t^{(r)}) \in \mathbb{R}^{768}, r \in \{L, R, F\}. \quad (2)$$

The frozen backbone supplies the image-level visual prior without receiving SSL gradient. The per-video tensor $E^{(r)} \in \mathbb{R}^{T \times 768}$ is computed once and cached in float16, trading per-iteration image-encoder forwards for an order-of-magnitude SSL speedup (cached vs. live in §4.1).

3.2 Self-Supervised Training of SignDino

Each stream $r \in \{L, R, F\}$ trains an *independent* SSL model with the DINOv3 recipe transposed to the time axis (Figure 4). The student is a temporal Transformer $f_\phi^{(r)}$ and the teacher its EMA copy $f_\phi^{(r)}$; two MLP heads $h_\psi^{\text{cls}}, h_\psi^{\text{frm}}$ map the CLS and the per-frame tokens to K prototypes, with EMA copies $h_\psi^{\text{cls}}, h_\psi^{\text{frm}}$ on the teacher side. The backbone from Eq. (2) stays frozen; only the temporal encoders and heads receive gradient. We reserve x for image crops and video input, f for the temporal encoders, and g, ℓ for global and local *temporal* crops; numeric constants are collected in the hyperparameter card of Table 5 and are repeated in the text only where they carry the argument. The stream index (r) is dropped below, as the three streams are trained identically and independently.

Temporal Transformer. f_ϕ projects the 768-d frozen embeddings to $D=384$, prepends a learnable temporal CLS token, adds a sinusoidal positional encoding indexed by absolute frame number, and applies $L=6$ pre-LN Transformer blocks. The validity mask $m^{(r)}$ propagates into self-attention so detector-missed frames are not attended. The CLS token feeds the crop-level distillation loss; per-frame tokens feed iBOT, Gram anchoring, and all downstream sequence heads.

Multi-temporal-crop sampling. Each iteration draws $N_g=2$ *global* and $N_l=8$ *local* temporal crops per video, mirroring the multi-crop recipe DINOv2/DINOv3 apply in space (Oquab et al., 2024; Siméoni et al., 2025). A temporal crop is a contiguous block of frames whose length is drawn independently per crop, $T_g \sim \mathcal{U}\{64, \dots, 96\}$ for globals and $T_l \sim \mathcal{U}\{10, \dots, 32\}$ for locals, with a uniformly sampled start frame. This is the temporal analogue of a multi-scale RandomResizedCrop: it exposes the student to varied clip durations

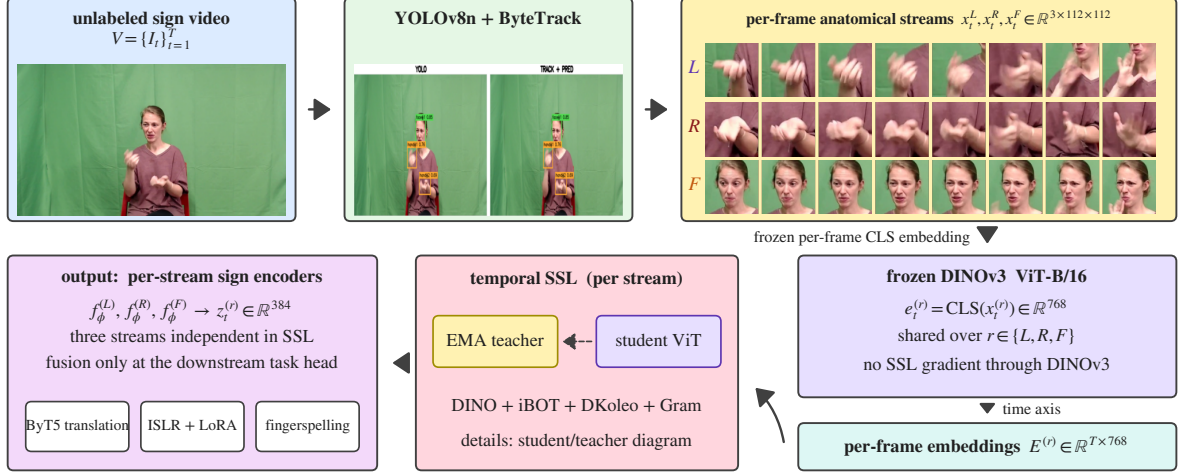


Figure 2: SIGNDINO pipeline. *Row A* (left-to-right): each source video V passes through a YOLOv8n+ByteTrack detector-tracker yielding synchronised per-frame anatomical streams x_t^L, x_t^R, x_t^F . *Row B* (right-to-left, snake): a frozen DINOv3 ViT-B/16 encodes every crop into a 768-d CLS token, producing three per-frame embedding sequences $E^{(r)} \in \mathbb{R}^{T \times 768}$, each feeding one independent temporal SSL trainer. *Row C*: stage-2 SignDino teachers serve as per-stream sign encoders $f_\phi^{(r)}$ that downstream translation, ISLR and fingerspelling heads consume as fused per-frame features. Figure 4 details the student/teacher loss.



Figure 3: Detector-first pipeline recovers hands YOLOv8n alone misses. Each cell shows the same OpenASL frame under YOLOv8n only (top) and YOLOv8n + ByteTrack with Kalman fill (bottom). Green boxes are active YOLO detections; magenta are tracker predictions for hands YOLOv8n missed at this frame.

rather than one fixed scale (ablated in Table 18). The teacher sees only the two globals; the student sees all ten crops, and inside each student global a random fraction of frame positions has its input embedding replaced by a learnable [MASK] token, supplying the masked positions for the iBOT term. Because crop lengths vary within a batch, collation right-pads each view to the batch maximum, and the validity mask removes padded positions from attention, from the iBOT mask set, and from the Gram loss.

Stage 1: SSL loss. The Stage-1 objective is the DINOv3 objective evaluated on temporal rather than spatial crops. It combines three terms, each defined below: a crop-level cross-entropy $\mathcal{L}_{\text{DINO}}$ between the student’s and the teacher’s prototype assignments of the CLS token; a per-frame masked cross-entropy $\mathcal{L}_{\text{iBOT}}$, the temporal counterpart of the masked-patch prediction objective introduced by iBOT (Zhou et al., 2022), which asks the student to predict the teacher’s prototype assignment at frames the student never saw; and $\mathcal{L}_{\text{DKoleo}}$, the distributed variant (Oquab et al., 2024) of the KoLeo differential-entropy regulariser (Sablayrolles et al., 2019), which spreads embeddings over the unit sphere so that they do not collapse onto a few directions:

$$\mathcal{L}_{\text{S1}} = \mathcal{L}_{\text{DINO}} + \mathcal{L}_{\text{iBOT}} + 0.1 \cdot \mathcal{L}_{\text{DKoleo}}. \quad (3)$$

Write ℓ for a student temporal crop and g for a teacher global temporal crop. Their CLS prototype distributions over the K prototypes are

$$p_T(g) = \text{SK}(h_\psi^{\text{cls}}(f_\phi(g))/\tau_T), \quad (4)$$

$$p_S(\ell) = \text{softmax}(h_\psi^{\text{cls}}(f_\phi(\ell))/\tau_S), \quad (5)$$

with teacher and student temperatures τ_T, τ_S . Here $\text{SK}(\cdot)$ is Sinkhorn–Knopp normalisation (Caron et al., 2020): a few iterations of alternating row and column normalisation of the batch of teacher

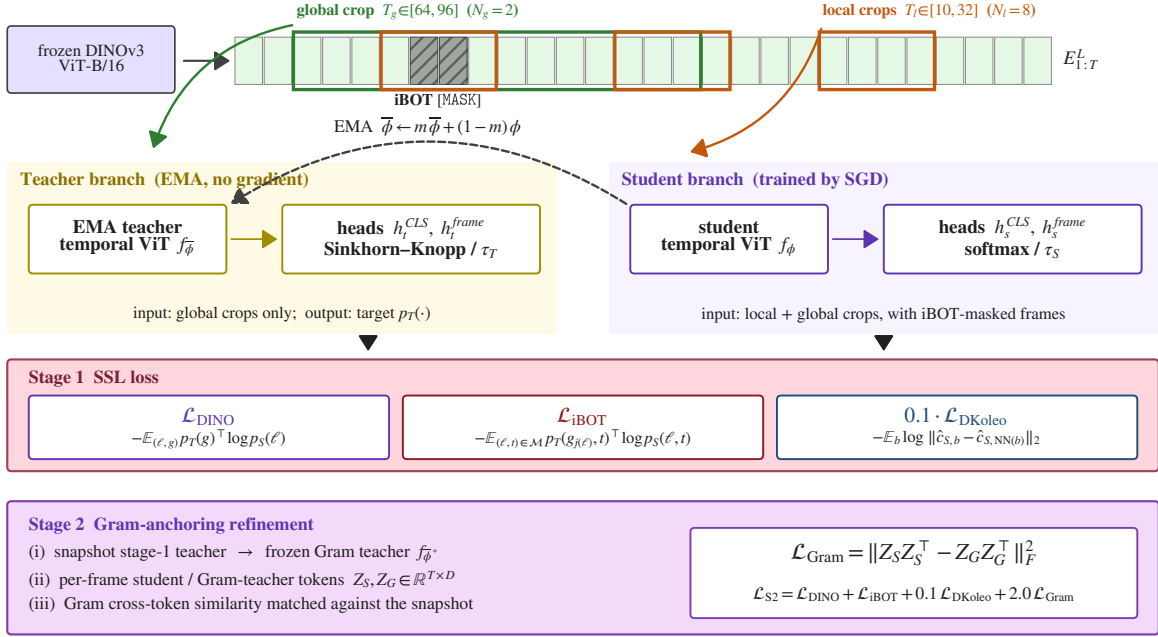


Figure 4: Temporal student/teacher game for one anatomical stream (shown for L). The frozen DINOv3 ViT-B/16 supplies per-frame embeddings $E_{1:T}^L$ once; the temporal SSL game runs on top. The teacher sees only global temporal crops of length $T_g \sim \mathcal{U}\{64, \dots, 96\}$, while the student sees both global and local crops of length $T_l \sim \mathcal{U}\{10, \dots, 32\}$ with a uniform fraction of frame positions replaced by the iBOT [MASK] token. The teacher is the EMA of the student. The Stage 1 loss combines three terms; Stage 2 adds a Gram anchoring term constraining frame-to-frame structure inside each clip.

scores, which forces the assignment to be near-uniform over the K prototypes across the batch and so prevents the teacher from routing every crop to the same prototype. It replaces the batch centring of DINOv2, following DINOv3. The CLS-distillation term averages over the set $\mathcal{P}$ of (student-crop, teacher-crop) pairs, excluding the global-global diagonal:

$$\mathcal{L}_{\text{DINO}} = -\mathbb{E}_{(\ell, g) \in \mathcal{P}} [p_T(g)^\top \log p_S(\ell)]. \quad (6)$$

The iBOT term applies the same cross-entropy at every masked frame position through the frame-token heads, where $p_T(g, t)$ and $p_S(\ell, t)$ are the prototype distributions of the frame- t token under h_{ψ}^{frm} and h_{ψ}^{frm} , $g_{j(\ell)}$ is the teacher global crop paired with student crop ℓ , and $\mathcal{M}$ is the set of (crop, frame) positions masked in the batch:

$$\mathcal{L}_{\text{iBOT}} = -\mathbb{E}_{(\ell, t) \in \mathcal{M}} [p_T(g_{j(\ell)}, t)^\top \log p_S(\ell, t)]. \quad (7)$$

Eq. (7) is an unweighted mean over $\mathcal{M}$ across the batch (per-mask, not per-crop), as in the original DINOv2/iBOT (Oquab et al., 2024; Zhou et al., 2022) formulation; padded positions never enter $\mathcal{M}$. The DKoleo term acts on the student CLS embeddings of a batch of B videos. Let $\hat{c}_{S,b}$ be the

L2-normalised student CLS embedding of the b -th video and $\text{NN}(b) = \arg \min_{b' \neq b} \|\hat{c}_{S,b} - \hat{c}_{S,b'}\|_2$ the index of its nearest neighbour in the same batch; maximising the log-distance to that nearest neighbour keeps the batch from contracting to a few points:

$$\mathcal{L}_{\text{DKoleo}} = -\mathbb{E}_b [\log \|\hat{c}_{S,b} - \hat{c}_{S,\text{NN}(b)}\|_2]. \quad (8)$$

Stage 2: Gram-anchoring refinement. After Stage-1 convergence we snapshot the EMA teacher as a frozen *Gram teacher* f_{ϕ^*} and add a frame-to-frame Gram loss. Let $Z_S, Z_G \in \mathbb{R}^{T \times D}$ collect the L2-normalised per-frame output tokens that the student f_ϕ and the Gram teacher f_{ϕ^*} produce on the same global temporal crop, one row per frame. Then $Z_S Z_S^\top$ and $Z_G Z_G^\top$ are their frame-to-frame cosine-similarity matrices, and the Gram loss matches them:

$$\mathcal{L}_{\text{Gram}} = \|Z_S Z_S^\top - Z_G Z_G^\top\|_F^2. \quad (9)$$

The stage-2 total loss is

$$\mathcal{L}_{\text{S2}} = \mathcal{L}_{\text{DINO}} + \mathcal{L}_{\text{iBOT}} + 0.1 \cdot \mathcal{L}_{\text{DKoleo}} + 2.0 \cdot \mathcal{L}_{\text{Gram}}. \quad (10)$$

Stage 1 drives per-frame tokens toward the crop-level CLS, while Gram anchoring preserves which

frames are similar versus dissimilar inside a clip — the temporal grammar of a sign (Bellugi and Fischer, 1972) that the downstream fusion head needs.

Where stream fusion happens. Per-stream tokens from every Transformer layer feed a learned softmax over the $L+1$ layers (Yang et al., 2021), giving a layer-weighted per-frame representation LayerNorm-projected to 256-d. The three streams are concatenated to 768-d and projected to the downstream input dimension (e.g., 1472 for ByT5-Base), following the SHUBERT §4.2 layer-weighted-sum design (their Table 6 reports ~ 6 BLEU over last-layer-only).

4 Experiments

Our protocol isolates the SSL objective under SHUBERT’s public-data setting: matching its source-corpus accounting, downstream datasets, task heads, optimisation schedules, decoding setup, and metric conventions wherever possible, we vary only the upstream representation, replacing SHUBERT’s clustered multi-stream hidden-unit encoder with our temporal-axis DINO encoder. Appendix E gives split, preprocessing, and metric details; Appendix C adds SHUBERT’s phonological-feature diagnostic. All experiments use ASL or ASL-to-English benchmarks.

4.1 Pre-Training Setup

Datasets. We pre-train SIGNDINO on the same public YouTube-ASL source split as SHUBERT (Uthus et al., 2023; Gueuwou et al., 2025b): clips intersecting the OpenASL evaluation set are removed and replaced with non-overlapping ASL videos from YouTube-SL-25 (Tanzer and Zhang, 2024), yielding ≈ 984 unique source hours. We apply our detector-first crop pipeline and temporal SSL objective to this decontaminated split; pre-training-hour figures follow SHUBERT’s public-data accounting.

Model. Per stream: $1 \times 86\text{M}$ frozen DINOv3 ViT-B/16 ($d=768$, FFN 3072) + $1 \times 3.5\text{M}$ trainable temporal Transformer ($L=6$, $D=384$, 6 heads) + $2 \times 2.5\text{M}$ DINO/iBOT projection heads to $K=8192$ prototypes. The frozen DINOv3 is shared across the three streams, so only $\approx 30\text{M}$ parameters are SSL-trained. Appendix E (Table 7) reports the two-stage SSL schedule.

Training compute. All SSL runs use $8 \times \text{NVIDIA A100 } 80\text{GB GPUs}$, training the three per-stream encoders sequentially on the same pool. Per-stage GPU-hours, wall-clock, and the one-off embedding-cache cost are tabulated in Appendix F.4 (Table 12); with the cache enabled, SSL training runs $\approx 50\times$ faster per step than the live regime.

4.2 Sign Language Translation

Architecture. The translation head replicates SHUBERT §4.2: per-frame stage-2 teacher features are LayerNorm + linear-projected, concatenated across streams, and projected to $d_{\text{model}}=1472$ as input embeddings for the ByT5-Base decoder (Xue et al., 2022), which generates UTF-8 byte tokens under cross-entropy with label-smoothing 0.2. Matching SHUBERT, we report quality as corpus BLEU-4 (Papineni et al., 2002) via SacreBLEU (Post, 2018) and BLEURT-20 (Selam et al., 2020); all “BLEU” entries are BLEU-4. The fusion projection (learned softmax over $L+1=7$ teacher layers, 256-d per stream, 768-d concatenated) is detailed in Appendix E.

Two-phase training. We follow SHUBERT’s two-stage SLT adaptation (hyperparameters in Appendix E, Table 8): the source stage trains on the decontaminated YouTube-ASL split (§4.1), the target stage fine-tunes on the original How2Sign or OpenASL split, and FLEURS-ASL is scored zero-shot after the source stage. The *live-fine-tune* variant updates the SignDino temporal encoders at $1/10$ the ByT5/head LR (DINOv3 frozen); the *frozen-features* variant trains only the fusion projection and ByT5. Table 1 reports the comparison.

Results. Live-fine-tuned SIGNDINO (Table 1) reaches $17.9/24.5$ BLEU on How2Sign/OpenASL, beating the strongest published SSL baseline SHUBERT (Gueuwou et al., 2025b) by $+1.7/+1.3$ BLEU and $+1.4/+1.1$ BLEURT. The frozen-feature variant (encoders fixed, only fusion projection + ByT5 trained) already surpasses SHUBERT (16.8 vs. 16.2 on H2S, 23.6 vs. 23.2 on OpenASL), so the temporal-axis features carry signal without encoder adaptation. Replacing the layer-weighted-sum head with the last layer alone lowers H2S to 15.8 (frozen-encoder counterpart 14.7 , Table 16), separating the pooling and fine-tuning contributions. On zero-shot FLEURS-ASL, SIGNDINO reaches 5.3 BLEU vs. SHUBERT 4.7 and YouTube-SL-25 (Tanzer and Zhang, 2024) 4.4 ,

Method	SSL	PT hrs	How2Sign		OpenASL		FLEURS-ASL
			BLEU↑	BLEURT↑	BLEU↑	BLEURT↑	BLEU↑
SSVP-SLT (Rust et al., 2024)	✓	1054	15.5	49.6	—	—	—
YouTube-ASL fine-tune (Uthus et al., 2023)	×	984	12.4	46.6	—	—	—
YouTube-SL-25 (Tanzer and Zhang, 2024)	×	3207	15.4	47.9	—	—	4.4
SignMusketeers (Gueuwou et al., 2025a)	✓	984	14.3	—	—	—	—
SpaMo (Hwang et al., 2025)	×	—	10.11	42.23	—	—	—
Uni-Sign (Li et al., 2025)	×	984	14.9	49.4	23.1	60.4	—
OpenASL (Shi et al., 2022)	×	—	—	—	6.7	31.1	—
SHUBERT (Gueuwou et al., 2025b)	✓	984	16.2	49.9	23.2	60.6	4.7
SIGNDINO							
SIGNDINO frozen + ByT5 (SHUBERT Tab. 7 “×”)	✓	984	16.8 $\pm$ 0.3	50.4 $\pm$ 0.4	23.6 $\pm$ 0.3	60.9 $\pm$ 0.3	4.9 $\pm$ 0.2
SIGNDINO live fine-tune (SHUBERT Tab. 7 “✓”)	✓	984	17.9 $\pm$ 0.3	51.3 $\pm$ 0.4	24.5 $\pm$ 0.3	61.7 $\pm$ 0.3	5.3 $\pm$ 0.2
SIGNDINO live, last-layer only (no weighted sum)	✓	984	15.8 $\pm$ 0.3	48.7 $\pm$ 0.4	23.0 $\pm$ 0.3	59.7 $\pm$ 0.4	4.6 $\pm$ 0.2

Table 1: Sign language translation on How2Sign, OpenASL and FLEURS-ASL under the SHUBERT public-data protocol. SSL = ✓ denotes self-supervised pre-training; PT hrs report source pre-training video hours used before benchmark-specific fine-tuning. For SIGNDINO, Phase 1 uses the OpenASL-decontaminated YouTube-ASL split, Phase 2 fine-tunes on How2Sign or OpenASL, and FLEURS-ASL is evaluated zero-shot. SpaMo reports How2Sign BLEU-4/BLEURT under its LLM-based gloss-free protocol and is included as a supervised LLM reference. The “frozen vs live” pair follows SHUBERT’s frozen/fine-tuned comparison. SIGNDINO entries are mean $\pm$ standard deviation over 3 random seeds; baselines are reported as published.

and exceeds the 580M-parameter supervised Uni-Sign (Li et al., 2025) by +1.4 BLEU on OpenASL at matched downstream capacity.

4.3 Isolated Sign Language Recognition

Task and architecture. We evaluate ISLR on ASL Citizen (Desai et al., 2023), Sem-Lex (Kezar et al., 2023), and WLASL2000 (Li et al., 2020), matching SHUBERT’s benchmark suite; as in SHUBERT, MSASL is excluded because its test set overlaps with the SSL source YouTube-ASL. The head follows SHUBERT §4.3 — stream-fused, mask-aware time-average + BatchNorm + Linear classifier under cross-entropy with label-smoothing 0.1. We report Recall@1/5/10 for ASL Citizen / Sem-Lex and per-instance/per-class top-1 accuracy for WLASL2000.

Adaptation regimes. We report three regimes: (i) frozen features + linear; (ii) rank-1 LoRA (Hu et al., 2022) on every Linear of the SignDino temporal encoders, matching SHUBERT §4.3’s 0.17M-parameter setting; (iii) full fine-tuning of the temporal encoders. Optimisation, schedule, and early-stopping rules are shared across regimes and reported in Appendix E. Table 2 gives the ISLR comparison alongside the fingerspelling-detection results discussed below.

Results. Under SHUBERT’s 0.17M-parameter rank-1 LoRA setting (Table 2), SIGNDINO improves Recall@1 by +0.02 on both ASL Citizen (0.67) and Sem-Lex (0.56); full fine-tuning lifts

these to 0.704 (+0.054) and 0.593 (+0.053). On WLASL2000 the same LoRA reaches 66.85/64.20 P-I/P-C, beating Uni-Sign (Li et al., 2025) (63.52/61.32) by +3.3 P-I at roughly one one-thousandth its trainable parameters; full fine-tuning reaches 68.45/65.92 (SHUBERT: 60.90/58.01). The frozen-feature variant stays competitive (0.63/0.52 R@1, comparable to I3D and SignCLIP) but does not exceed SHUBERT, so rank-1 LoRA is the operating point where temporal-axis features become consistently superior to the matched discrete-unit baseline. We compare only against public SSL/feature-based baselines evaluated under the SHUBERT suite.

4.4 Phonological Feature Probing

Beyond whole-sign classification, we ask whether SIGNDINO preserves sub-lexical structure. Following SHUBERT’s phonological benchmark, we train 16 feature-specific classifiers on Sem-Lex and ASL Citizen for ASL-Lex 2.0 attributes (hand-shape, selected fingers, location, contact, motion). This probe neither trains the representation nor informs model selection; it diagnoses whether temporal-axis SSL encodes linguistically meaningful sign-internal categories. Full protocol and per-feature Recall@1 are in Appendix C (Table 6).

4.5 Fingerspelling Detection

ASL-STEM-Wiki (Yin et al., 2024) provides long ASL videos with annotated fingerspelling intervals. Following the ASL-STEM-Wiki and SHU-

Method	Trainable	ASL Citizen			Sem-Lex			WLASL2000		ASL-STEM-Wiki
		R@1↑	R@5↑	R@10↑	R@1↑	R@5↑	R@10↑	P-I↑	P-C↑	mIoU↑
<i>Prior published results</i>										
ST-GCN (Desai et al., 2023)	0.45M	0.60	0.82	0.88	—	—	—	—	—	—
SignCLIP (Jiang et al., 2024)	217M	0.60	0.84	0.89	0.30	0.48	0.55	—	—	—
I3D (Desai et al., 2023)	25M	0.63	0.86	0.91	—	—	—	—	—	—
SignBERT+ (Hu et al., 2023)	—	—	—	—	—	—	—	48.85	46.37	—
MSLU (Zhou et al., 2024)	—	—	—	—	—	—	—	56.29	53.29	—
Uni-Sign (Li et al., 2025)	580M	—	—	—	—	—	—	63.52	61.32	—
SHUBERT (rank-1 LoRA) (Gueuwou et al., 2025b)	0.17M	0.65	0.87	0.91	0.54	0.74	0.80	60.90	58.01	0.40
SIGNDINO										
SIGNDINO frozen + linear	linear	0.63±.01	0.86±.01	0.91±.01	0.52±.01	0.72±.01	0.79±.01	60.1±0.4	57.5±0.4	0.39±.01
SIGNDINO rank-1 LoRA	0.17M	0.67±.01	0.88±.01	0.92±.01	0.56±.01	0.76±.01	0.82±.01	66.9±0.4	64.2±0.4	0.41±.01
SIGNDINO full fine-tune	all	0.704±.01	0.89±.01	0.93±.01	0.593±.01	0.78±.01	0.84±.01	68.5±0.4	65.9±0.4	0.43±.01

Table 2: Isolated sign recognition and fingerspelling detection under the SHUBERT evaluation suite. ASL Citizen and Sem-Lex use Recall@1/5/10; WLASL2000 reports per-instance (P-I) and per-class (P-C) top-1 accuracy; ASL-STEM-Wiki reports mean interval-IoU. Sem-Lex numbers follow the full-test-set protocol of SHUBERT, and are therefore not directly comparable to the original Sem-Lex report, which uses a reduced test set. SIGNDINO entries are mean $\pm$ standard deviation over 3 random seeds; baselines are reported as published.

BERT setup, we cast detection as per-frame binary classification: a two-layer Transformer atop the fused per-stream features produces frame logits, post-processed by thresholding ($\tau=0.5$), median smoothing (kernel 3), and contiguous interval extraction ($L_{\min}=3$ frames). Evaluation uses mean interval-IoU under the dataset cross-validation protocol; the ASL-STEM-Wiki mIoU column of Table 2 reports it alongside the ISLR benchmarks. Under rank-1 LoRA, SIGNDINO reaches mIoU 0.41 vs. SHUBERT 0.40, and full fine-tuning of the SignDino temporal encoder improves to 0.43, an absolute +0.03 gain over the strongest publicly comparable SSL baseline at the same downstream protocol. Appendix E summarises the exact protocol.

4.6 Ablations

The main body reports the three ablation groups that most directly test our contribution — per-stream contribution and the SIGNDINO-specific design choices (Gram anchoring, teacher window length, detector-first crops), with masking strategy summarised below. The remaining axes (data scale, layer pooling, frozen-vs-live, crop-length randomisation, visual backbone, decoder) are tabulated in Appendix F.6. All ablations share the contract of Appendix E; BLEU is corpus BLEU-4 on the *How2Sign* test split unless noted.

Masking strategy. The SignDino-native frame-only mask is the best of five strategies (BLEURT 50.4), ahead of SHUBERT’s random/time/channel masking (48.9/48.1/47.6) and of adding a stream-level mask (-0.7 BLEURT); we keep the frame-only default (Table 15, Appendix F.6).

Streams kept at fusion head	BLEU
LH + Face (no RH)	9.8
RH + Face (no LH)	11.2
LH + RH (no Face)	14.6
LH + RH + Face (default)	17.9

Table 3: Per-stream leave-one-out ablation, evaluated on the *How2Sign* test split with SIGNDINO live fine-tuned (same regime as the SIGNDINO live row of Table 1). Each row drops one of the three default streams (LH, RH, Face) at the fusion head while the per-stream SSL stage is unchanged; the bottom row reproduces the SIGNDINO live row of Table 1.

Stream contribution. Reported at the live-fine-tune operating point, so the default row reproduces the SIGNDINO live row of Table 1 (17.9 BLEU). Dropping the right-hand stream is most damaging (-8.1 BLEU), followed by the left hand (-6.7) and the face (-3.3), consistent with right-hand dominance in the predominantly right-handed YouTube-ASL/How2Sign signers. The two-handed manual channel carries most of the lexical signal; the face contributes a smaller but measurable mouthing / non-manual cue.

Ablations specific to SIGNDINO. We probe three choices unique to the temporal-axis recipe (Table 4). (i) *No Gram anchor* removes $\mathcal{L}_{\text{Gram}}$ from stage 2: H2S BLEU drops 17.9→16.5 (-1.4) and ASL Citizen R@1 0.704→0.67, so stage-2 frame-similarity preservation contributes beyond the stage-1 prototype losses. (ii) *Teacher short window* forces the teacher to draw crops from the local range $\mathcal{U}\{10, \dots, 32\}$ (the temporal analog of removing global crops in image DINO); this is the sharpest drop, -2.1 BLEU (15.8), confirming the longer global temporal context as the dominant supervisory signal. (iii) *YOLO only* disables ByteTrack and feeds raw per-frame detections, costing

Ablation	H2S BLEU	ASLC R@1
Full SIGNDINO	17.9	0.704
no Gram anchor (Eq. 9)	16.5	0.67
Teacher short window only	15.8	0.65
YOLO only (no ByteTrack)	16.4	0.61

Table 4: Ablations specific to SIGNDINO, evaluated on the *How2Sign test* split (H2S BLEU column) and the *ASL Citizen test* split (ASLC R@1 column). Each row toggles one factor and reuses the same downstream head; the “Full SIGNDINO” row reproduces the SIGNDINO live row of Table 1 (H2S BLEU) and the SIGNDINO full fine-tune row of Table 2 (ASLC R@1).

-1.5 BLEU (16.4) and -0.094 R@1 (0.61): unfilled single-frame detector failures yield a sparser, noisier crop sequence (cf. Fig. 3), with ISLR the more sensitive task.

Additional design-axis and protocol ablations.

Three further design axes (full tables in Appendix F.6) confirm robustness: the randomised crop-length sampler beats fixed (T_l, T_g) pairs by 0.4–0.6 BLEU including iso-FLOP controls (Table 18); H2S BLEU spans only 16.5–18.0 across five frozen DINO backbones, with ViT-B/16 the compute-aware default (Table 19); and SIGNDINO beats SHUBERT by +1.4–+1.7 BLEU across five decoders, with byte-level ByT5-Base the sweet spot (Table 20). The SHUBERT-protocol axes confirm pre-training is unsaturated (+3.3 BLEU, 10% → 100%), the layer-weighted-sum recovers +7.4 BLEU over raw DINOv3, and live fine-tuning adds +1.1 BLEU.

Training and downstream diagnostics. Three diagnostics confirm the SSL stays healthy and the fusion routes streams as intended: teacher prototype entropy stays above 0.91 (no collapse), per-task layer weights concentrate ISLR/FS on the deepest layers while SLT spreads wider, and the fusion head’s Face query attends 0.57 to itself for SLT (mouthings) but near-uniformly for fingerspelling. Full tables, a cluster analysis, and a cross-sentence t-SNE are in Appendices F, B and D.

5 Conclusion

We presented SIGNDINO, a self-supervised sign-video representation learner that transposes the DINOv3 teacher–student recipe onto the temporal axis of tracked ASL articulator streams (left/right hand and face), trained with DINO + iBOT + DKoleo plus a Stage-2 Gram-anchoring objective on a frozen DINOv3 + YOLOv8n–ByteTrack

front end. Under the SHUBERT public-data protocol, SIGNDINO improves over the strongest public SSL baseline across translation, ISLR, fingerspelling, and phonological probing, with a frozen-feature operating point that already exceeds the previous live-fine-tuned baseline on translation. We release the framework, code, and checkpoints as a strong public SSL baseline for sign-language video.

Limitations

SIGNDINO inherits several practical limitations. (i) The SSL signal flows through YOLOv8n + ByteTrack hand and face detections; long tracker dropouts under occlusion or fast motion corrupt the per-stream input, and the per-frame validity mask removes only the worst cases. (ii) The frozen DINOv3 backbone is trained on web images and not on signing video; a stream-specific fine-tune (as SHUBERT does for hand and face DINOv2) may close some of the gap and is left for future work. (iii) We use three streams (LH, RH, face) and do not explicitly model body-pose context; adding a 14-d upper-body channel as a fourth stream is a one-config-line change in the fusion module. (iv) All experiments are on ASL or ASL-to-English benchmarks; extending the temporal-axis objective to other sign languages and multilingual corpora remains future work, with the methodological-care, consent, and Deaf-community-review considerations of Desai et al. (2024). (v) The detector-tracker plus frozen image encoder introduce a non-trivial one-off preprocessing cost (Table 12); the per-stream temporal-SSL stages on top of the cached embeddings are modest by comparison, and the full upstream budget remains below the upstream cost of the strongest publicly comparable SSL baseline (Appendix F.4).

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A Default Hyperparameter Card

Table 5 is the high-level configuration card; the detailed two-stage SSL pre-training schedule (Table 7) and the source/target translation training schedule (Table 8) are reported under Appendix E.

Hyperparameter	Default value
Backbone	frozen DINOv3 ViT-B/16
Streams	LH, RH, face (3 streams, independent SSL)
Temporal encoder	$L=6$, $D=384$, 6 heads, MLP ratio 4
Crops (contiguous)	$T_g \sim \mathcal{U}\{64-96\}$, $T_l \sim \mathcal{U}\{10-32\}$; $N_g=2$, $N_l=8$
Temperatures	$\tau_T : 0.04 \rightarrow 0.07$ (30 ep), $\tau_S = 0.1$
EMA momentum	cosine from 0.994 to 1.0
Prototype K	8192
Gram weight (stage 2)	2.0
DKoleo weight	0.1
Optimiser	AdamW, cosine decay, bf16
Detector + tracker	YOLOv8n + ByteTrack, gaps of $\leq K=3$ frames interpolated
Downstream SLT	ByT5-Base, beam = 5, $L_{\max}=384$, label smoothing 0.2
Downstream ISLR	BN + linear; rank-1 LoRA; 125 epochs, batch 128
Downstream FS	2-layer Transformer + frame head; threshold 0.5

Table 5: Default hyperparameter card. Values summarise the default configuration used for the main experiments and ablations.

EMA momentum schedule under variable-length crops. We retain the per-step EMA momentum schedule of DINOv3 ($m : 0.994 \rightarrow 1.0$ on a cosine over the full training run) unchanged, despite the fact that our variable-length crop sampler exposes a stochastic per-step token count $N_{\text{tok}} = N_g T_g + N_l T_l \in [208, 448]$ in place of DINO/DINOv2/DINOv3’s fixed $N_g \cdot 256 + N_l \cdot 36$ ViT-patch budget (RandomResizedCrop in image DINO varies the source-image scale but resizes every crop to a fixed output resolution, so the per-step ViT-token count is constant by construction). The schedule transfers in the temporal setting because all three SSL losses — DINO, iBOT, and DKoleo — are computed as averages rather than sums: the DINO cross-entropy averages over a fixed count of (ℓ, g) crop pairs (§3.2), the iBOT cross-entropy averages over the per-step mask set $\mathcal{M}$ with the per-mask normalisation discussed after Eq. (7), and the DKoleo regulariser averages over a fixed-count set of student CLS tokens. Per-step gradient magnitude is therefore approximately invariant to the realised N_{tok} , and the EMA momentum schedule — which tracks per-step parameter movement rather than per-token information flow — inherits its DINOv3 calibration directly without recalibration.

B Qualitative Cluster Analysis of SIGNDINO Features

To probe what the SIGNDINO encoder actually captures, we run a fully zero-shot qualitative analysis on held-out frames, mirroring the cluster-visualisation protocol of SHUBERT Appendix B (Figures 5–7 there) (Gueuwou et al., 2025b). After the SSL training described in §3.2 — no labels of any kind, and with OpenASL clips that overlap with the YouTube-ASL pre-training source already decontaminated at the SSL stage (§4.1) — we apply the frozen SIGNDINO pipeline to a fresh sample of *never-trained-on* How2Sign and OpenASL videos: YOLOv8n + ByteTrack extracts synchronised face / left-hand / right-hand crops, each crop is embedded through the frozen DINOv3 ViT-B/16 (§3.1), and the per-frame CLS vectors are L2-normalised. We then run k -means ($K=250$ for the face stream, $K=200$ for each hand stream) on these embeddings and select rows by a farthest-first traversal in centroid space, filtered to clusters whose members span a sufficiently diverse set of source videos (so single-signer / single-shot clusters are excluded). For each chosen cluster we render the 10 nearest-to-centroid crops. The procedure is intended to be *descriptive*: we do not use the cluster labels in any downstream evaluation. Figures 5–7 show the result. Notably, the face stream surfaces clusters defined by *expression and pose* (hand-on-face mouthings, side-profile, neutral closed mouth) that recur across multiple identities — analogous to SHUBERT cluster 141 — and each hand stream surfaces *handshape* clusters (pointing / spread-fingers / thumb-up / two-hand contact) that recur across many backgrounds and skin tones, evidence that the temporal-axis SIGNDINO encoder is internalising sub-lexical articulator structure rather than identity or background. Anatomical hand labels follow the signer’s perspective: “left hand” (Figure 6) corresponds to image-right under our convention, since signers face the camera.

C Phonological Feature Recognition

Motivation and protocol. Following SHUBERT Appendix B/C, we include a phonological-feature recognition benchmark in addition to task-level evaluation. This probe asks whether a learned video representation preserves the sub-lexical structure from which ASL signs are formed, rather than only supporting a whole-sign class label or an En-

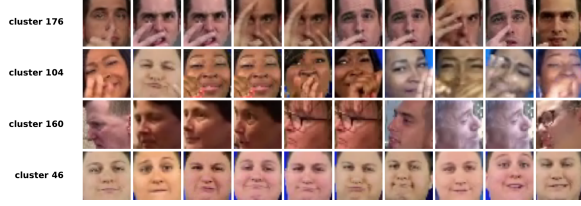


Figure 5: Sample *face* clusters from SIGNDINO features on held-out How2Sign and OpenASL frames. Four k -means clusters ($K=250$); each row shows 10 nearest-to-centroid members. Cluster 176 captures “hand-on-face mouthings” across multiple identities; cluster 104 captures hand-near-mouth gestures with varied skin tones; cluster 160 surfaces side-profile / looking-down poses; cluster 46 collects neutral closed-mouth frontals.

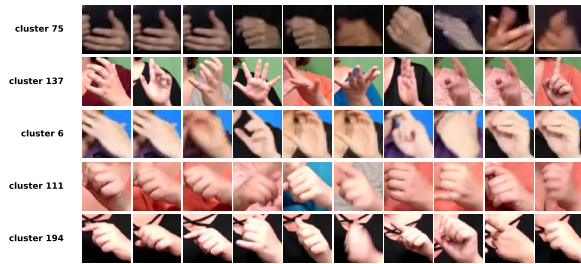


Figure 6: Sample *left-hand* clusters from SIGNDINO features on held-out How2Sign and OpenASL frames. Five k -means clusters ($K=200$). Cluster 75 captures thumb-up / closed-fist handshapes; cluster 137 captures the spread-fingers (“5”) handshape across multiple backgrounds; clusters 6 and 111 capture mid-motion signing gestures; cluster 194 captures two-hand contact poses.

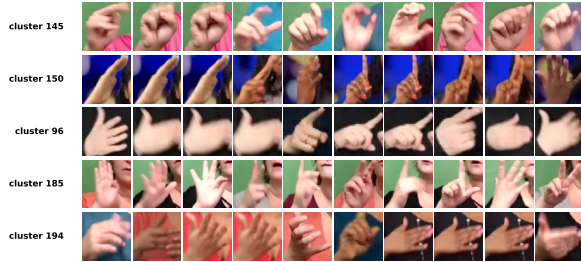


Figure 7: Sample *right-hand* clusters from SIGNDINO features on held-out How2Sign and OpenASL frames. Five k -means clusters ($K=200$). Cluster 145 captures pointing / extended-index handshapes across multiple backgrounds; cluster 150 captures a related forward-pointing pose on dark backgrounds; cluster 96 captures the thumb-up / closed-fist family; cluster 185 captures open-palm / spread-fingers gestures; cluster 194 captures motion-blurred active gesturing.

glish translation. We adopt this benchmark because it is a direct diagnostic of whether the model has encoded deeper sign-language information: location, handshape, selected fingers, contact, and motion are not incidental visual attributes, but lin-

guistically meaningful components of ASL signs. If a representation supports accurate prediction of these properties across datasets, it provides stronger evidence that the SSL model has captured sign-internal structure and category information.

We use the same two datasets and feature inventory as SHUBERT: Sem-Lex (Kezar et al., 2023) and ASL Citizen (Desai et al., 2023), with phonological labels taken from ASL-Lex 2.0 (Sehyr et al., 2021). For each isolated-sign video, the stream-fused SIGNDINO representation is temporally averaged with the attention mask and passed to 16 parallel linear classification heads, one per phonological feature. The training setup follows the ISLR recipe in §4.3, except that the single gloss classifier is replaced by feature-specific heads and weight decay is removed, matching the SHUBERT probe setting. We report Recall@1, i.e., top-1 accuracy, separately for each feature.

Phonology results. Averaged over the 16 phonological features, SIGNDINO reaches 87.64 R@1 on Sem-Lex and 91.85 R@1 on ASL Citizen, compared to SHUBERT’s 79.38 and 85.02, a matched-protocol gain of +8.26 and +6.83 R@1 respectively. The gains are largest on temporally articulated features — Path Movement (+8.37/+8.30), Wrist Twist (+6.53/+6.50), Repeated Movement (+4.35/+6.28) — and on fine handshape categories — Selected Fingers (+10.41/+4.23), Flexion (+6.10/+5.10), Handshape (+8.23/+8.71). On the categorical features that were already strong for SHUBERT, SIGNDINO also closes most of the remaining headroom: Major Location reaches 94.43/94.96 R@1 (+9.66/+4.74) and Sign Type reaches 90.72/99.50 (+6.08/+7.96). The consistent gain across all 16 features — with no feature regressing — suggests that the temporal student-teacher distillation preserves substantially more sub-lexical structure than the discrete hidden-unit objective used by SHUBERT.

To the best of our knowledge, SHUBERT was the first work to report phonological-feature recognition accuracies on ASL Citizen under this setting, and the Sem-Lex numbers are not directly comparable to the original Sem-Lex report because SHUBERT evaluates on the entire public test set. We therefore follow SHUBERT and treat this benchmark as a reusable diagnostic for future sign representation learning work. In our setting, it is especially useful because SIGNDINO replaces discrete hidden-unit prediction with temporal student-

Phonological feature	Method	Sem-Lex R@1	ASL Citizen R@1
Major Location	SHUBERT	84.77	90.22
	SIGNDINO	94.43	94.96
Minor Location	SHUBERT	71.30	80.00
	SIGNDINO	76.83	85.79
Second Minor Location	SHUBERT	73.28	81.18
	SIGNDINO	82.34	90.73
Contact	SHUBERT	86.84	91.57
	SIGNDINO	98.93	96.94
Thumb Contact	SHUBERT	84.74	87.52
	SIGNDINO	92.55	92.16
Sign Type	SHUBERT	84.64	91.54
	SIGNDINO	90.72	99.50
Repeated Movement	SHUBERT	82.65	89.93
	SIGNDINO	87.00	96.21
Path Movement	SHUBERT	72.75	79.42
	SIGNDINO	81.12	87.72
Wrist Twist	SHUBERT	90.58	93.00
	SIGNDINO	97.11	99.50
Selected Fingers	SHUBERT	79.53	83.44
	SIGNDINO	89.94	87.67
Thumb Position	SHUBERT	86.04	88.19
	SIGNDINO	97.28	98.76
Flexion	SHUBERT	72.64	77.73
	SIGNDINO	78.74	82.83
Spread	SHUBERT	79.42	84.80
	SIGNDINO	90.99	91.89
Spread Change	SHUBERT	81.60	86.58
	SIGNDINO	86.44	91.75
Nondominant Handshape	SHUBERT	76.32	84.32
	SIGNDINO	86.60	93.63
Handshape	SHUBERT	62.93	70.80
	SIGNDINO	71.16	79.51
Average	SHUBERT	79.38	85.02
	SIGNDINO	87.64	91.85

Table 6: Phonological feature recognition on Sem-Lex and ASL Citizen. The table follows the SHUBERT Appendix B layout; R@1 is top-1 accuracy for each feature-specific classifier. For every feature we report the SHUBERT baseline above and SIGNDINO below; SIGNDINO rows are bolded to mark the matched-protocol gain.

teacher distillation; the probe tests whether this continuous temporal objective still preserves the linguistic categories encoded by ASL-Lex. The full feature vocabulary used by the 16 classification heads, together with a per-dimension reading of the gains, is listed in Appendix I.

D Cross-Sentence t-SNE of Translation-Trained Features

After source-stage + target-stage translation training (§4.2, How2Sign target), the SignDino temporal encoders together with the stream-fusion projection are live-fine-tuned end-to-end with ByT5. We then take the resulting encoders — frozen

post-training — and run a cross-sentence t-SNE on per-word feature aggregates extracted on a 2400-sentence sample drawn from the union of train, validation and test splits of How2Sign and the three ISLR datasets (ASL Citizen, Sem-Lex, WLASL2000). For each occurrence of a chosen English target word in the sentence-level annotation, we slice out the time-aligned signing window, average the per-frame fused stream representation over that window, and project all word-occurrence vectors jointly with t-SNE (perplexity 30, 1000 iterations). Per-occurrence points for the chosen target words are coloured; all other vocabulary occurrences are plotted in light grey as background.

This visualisation is a probe of the alignment between the trained SIGNDINO representation and English-token semantics: if the SLT-tuned encoder has internalised the alignment that the downstream translation loss imposes, sign occurrences of words with similar English meaning should land in nearby regions of the embedding, and sign occurrences of words with distinct meaning should land in distant regions. We show two lexical neighbourhoods — one of nouns and one of verbs — in Figure 8. Panel (a) takes a noun neighbourhood: five family-relation nouns (“mother”, “father”, “brother”, “sister”, “child”) concentrate in a dense central region of the projection, while two weather nouns (“rain”, “snow”) sit far from the family-noun cluster and also far from each other. Panel (b) takes a verb neighbourhood: three visual-perception verbs (“see”, “look”, “watch”) form a tight right-side cluster; three motion verbs (“walk”, “run”, “jump”) form a tight left-side cluster; the action verb “sleep” — belonging to neither perception nor locomotion — sits in its own region between them. The visualisation is descriptive — it is not used for any model selection or quantitative claim — but it provides a direct, sentence-level read on the kind of semantic structure that emerges after the SLT objective is applied on top of the SSL representation.

E Reproducibility Checklist

This appendix records the experimental contract needed to reproduce the results without exposing repository-specific file names. The released artefacts will include the trained per-stream SIGNDINO checkpoints, extracted features for the evaluated benchmarks, and configuration files corresponding to the hyperparameters reported in Tables 7, 8, and 5.

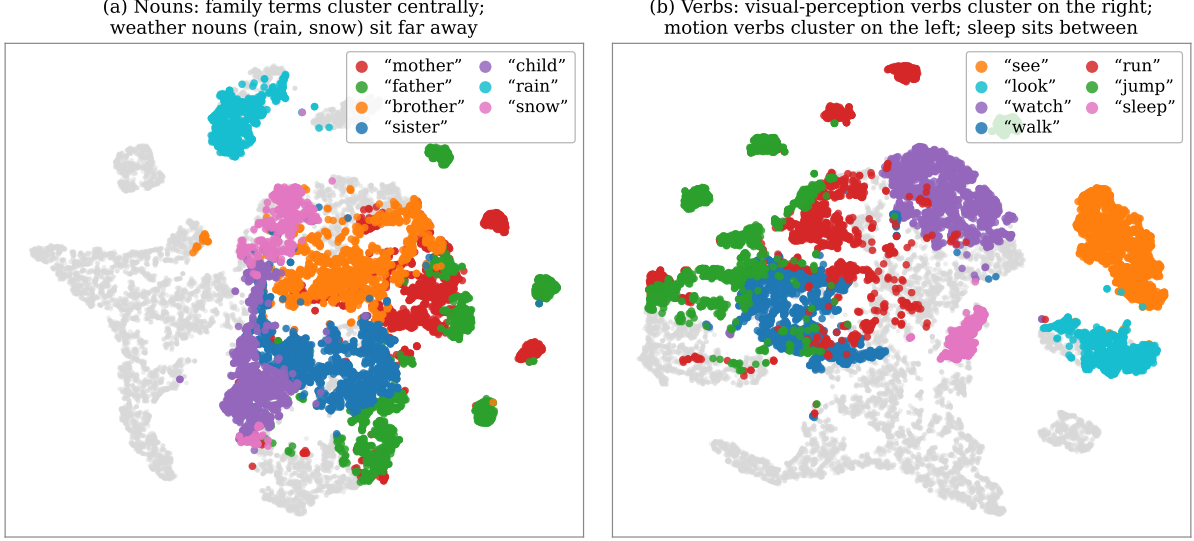


Figure 8: Cross-sentence t-SNE of SIGNDINO features after SLT fine-tuning, on a 2400-sentence sample drawn from How2Sign and the three ISLR datasets. Each coloured point is one in-sentence occurrence of the labelled English target word, projected by t-SNE on the post-SLT SIGNDINO stream-fused per-window features. Grey points are all other vocabulary occurrences in the same sample (background). (a) Common nouns: family terms (mother/father/brother/sister/child) concentrate in a central region; weather nouns (rain, snow) sit far from the family cluster. (b) Common verbs: visual-perception verbs (see/look/watch) cluster on the right; motion verbs (walk/run/jump) cluster on the left; “sleep” occupies a distinct region between the two verb groups.

Hyperparameter	Stage 1 (DINO+iBOT+DKoleo)	Stage 2 (Gram refine)
Steps	400K	50K
Batch (videos / GPU)	16	16
Optimiser	AdamW	AdamW
Base LR	5×10^{-4}	2×10^{-4}
LR schedule	5% warmup + cosine	1% warmup + cosine
Weight decay	$0.04 \rightarrow 0.4$	$0.04 \rightarrow 0.4$
EMA momentum	$0.994 \rightarrow 1.0$	$0.999 \rightarrow 1.0$
Teacher τ_T	$0.04 \rightarrow 0.07$ (30ep)	0.07
Student τ_S	0.1	0.1
Multi-temporal crops	$N_g=2, N_l=8$	same
T_g/T_l ranges (contiguous)	$\mathcal{U}\{64-96\} / \mathcal{U}\{10-32\}$	same
iBOT mask ratio	25–50%	same
Loss weights	Eq. (3)	Eq. (10)
Precision	bf16 autocast	bf16 autocast

Table 7: SIGNDINO pre-training hyperparameters for stage 1 and stage 2.

Source split and decontamination. For self-supervised pre-training and source-stage SLT training, we follow the SHUBERT public-data protocol. The source corpus is YouTube-ASL (Uthus et al., 2023). Clips that intersect the OpenASL evaluation set (Shi et al., 2022) are removed to avoid test contamination, and the removed duration is replaced with non-overlapping ASL videos from YouTube-SL-25 (Tanzer and Zhang, 2024). This keeps the public source corpus at approximately 984 source hours. We also do not report MSASL because its test set overlaps with YouTube-ASL, following the

	Source stage	Target stage
Data	decontaminated YouTube-ASL	How2Sign / OpenASL train
Steps	250K	50K
Optimiser	AdamW	AdamW
	$(\beta_1=0.9, \beta_2=0.99)$	
LR ByT5 + head	5×10^{-4}	1×10^{-4}
LR SignDINO (live)	5×10^{-5}	1×10^{-5}
Warmup	10K cosine	5K cosine
Effective batch	16 utt. (2 / GPU $\times$ accum 8)	16
Weight decay	0.1	0.1
Label smoothing	0.2	0.2
Decoding	beam = 5, $L_{\max}=384$	beam = 5
Hardware	8 $\times$ A100 80GB	8 $\times$ A100 80GB

Table 8: Translation training hyperparameters. The source stage follows SHUBERT §4.2; the target stage fine-tunes on the original How2Sign/OpenASL training split and evaluates on the corresponding validation/test splits.

same rationale as SHUBERT.

Data and preprocessing. For source pre-training, each YouTube-ASL video is passed through YOLOv8n + ByteTrack to obtain synchronised left-hand, right-hand, and face crop streams. For target-stage fine-tuning, How2Sign and OpenASL videos are processed once from the original training split, matching the main SHUBERT-style protocol. Validation and test videos are also processed once. Each stream carries a binary valid-

ity mask indicating detector failures. Short detector dropouts are linearly interpolated at the track level; longer failures are excluded from SSL losses through the validity mask. Crops are stored at 112×112 and resized to 224×224 for the frozen DINOv3 encoder. In result tables, pre-training hours are always reported as unique source-video hours, matching SHUBERT’s accounting.

Embedding cache and SSL training. Each crop stream is encoded once by the frozen DINOv3 ViT-B/16 into a float16 tensor $E^{(r)} \in \mathbb{R}^{T \times 768}$. Stage 1 trains the temporal student and EMA teacher for 400K steps with the DINO+iBOT+DKoleo objective in Eq. (3); Stage 2 refines the model for 50K steps with the additional Gram loss in Eq. (10). The default crop sampler draws two global windows of length $T_g \sim \mathcal{U}\{64-96\}$ frames and eight local windows of length $T_l \sim \mathcal{U}\{10-32\}$ frames, each as a contiguous block with an independently drawn start frame (§3.2), and applies an iBOT frame-mask ratio between 25% and 50%.

Translation. Translation follows SHUBERT’s source-to-target adaptation protocol. The source stage trains the fusion projection and ByT5-Base translation model on the decontaminated YouTube-ASL split for 250K steps. The target stage fine-tunes on the original training split of How2Sign (Duarte et al., 2021) or OpenASL (Shi et al., 2022) for 50K steps. FLEURS-ASL (Tanzer, 2024b) is evaluated zero-shot, since it has no training split in this protocol. Decoding uses beam search with width 5 and maximum output length 384. We report BLEU-4 (Papineni et al., 2002) computed with SacreBLEU (Post, 2018) and BLEURT (Sellam et al., 2020) computed with the BLEURT-20 checkpoint, following SHUBERT.

ISLR, fingerspelling, and phonology. ISLR evaluates ASL Citizen, Sem-Lex, and WLASL2000. The stream-fused representation is time-averaged with the attention mask, passed through BatchNorm and a linear classifier, and adapted either as frozen features, rank-1 LoRA, or full fine-tuning as described in §4.3. We report Recall@1/5/10 for ASL Citizen and Sem-Lex, and per-instance/per-class top-1 accuracy for WLASL2000. Fingerspelling detection on ASL-STEM-Wiki is trained as per-frame binary classification and evaluated by mean interval-IoU under the dataset cross-validation protocol. The phonological-feature probe in Appendix C uses 16

simultaneous feature-specific heads and reports Recall@1 for each ASL-Lex feature.

Compute, software, and seeds. The SSL stages use bf16 autocast and AdamW. The default pre-training run uses $8 \times$ A100 80 GB GPUs, with the three streams trained sequentially on the same 8-GPU pool; Stage 1 requires approximately 240 GPU-hours per stream and Stage 2 adds approximately 40 GPU-hours per stream (Appendix F.4, Table 12). With the embedding cache enabled, the temporal SSL updates are substantially faster because the frozen image backbone is not evaluated inside the training loop. Experiments are run with PyTorch 2.6, CUDA 12.4, HuggingFace Transformers for ByT5, SacreBLEU for BLEU, and BLEURT-20 for BLEURT. Unless otherwise stated, all reported runs use seed 0; ablations use the same data splits, crop streams, and downstream schedules as the corresponding main results.

Responsible use. Sign-language video contains identifiable signer information. We therefore retain the privacy-aware face processing options described by prior sign-language work (Rust et al., 2024; Desai et al., 2024), and we report preprocessing choices so that downstream users can audit which visual cues are retained.

F Training and Downstream Diagnostics

This appendix expands the three diagnostics summarised in §4.6 into full per-stream / per-task tables. The diagnostics probe (i) whether the temporal-axis SSL stays in the non-collapsed cluster regime during pre-training, (ii) which encoder depth each downstream task draws from, and (iii) whether the downstream fusion head attends across streams in a way that is consistent with linguistic prior knowledge about ASL articulators (face-cued mouthings, hand-cued lexical signs and fingerspelling).

F.1 Prototype entropy (cluster non-collapse)

Protocol. The teacher prototype distribution $p_{\text{teacher}} \in \Delta^{K-1}$ is the Sinkhorn–Knopp-normalised soft-assignment over $K=8192$ prototypes (§3.2). For each stream we periodically average p_{teacher} over a held-out batch of 1024 frames and report the normalised Shannon entropy $H(p_{\text{teacher}})/\log K \in [0, 1]$. A value near 1 means the teacher uses all prototypes near-uniformly (no collapse); a value near 0 means the teacher concentrates mass on a small subset of prototypes.

Following the DINO-family diagnostics convention (Caron et al., 2021; Siméoni et al., 2025), we use this normalised entropy as a non-collapse indicator: a value near 1 means the teacher distributes mass uniformly across all prototypes, while a sharp drop signals concentration of the teacher onto a small subset of prototypes.

Observed trajectory. Table 9 reports the normalised entropy at six checkpoints across the two-stage SSL schedule. Entropy decreases monotonically from 1.000 at initialisation to ≈ 0.92 on the hand streams and ≈ 0.94 on the face stream at Stage 2 convergence, all comfortably above the collapse regime. The face stream retains higher entropy than the hand streams throughout, consistent with a more diverse facial-expression vocabulary than hand configurations on YouTube-ASL.

SSL checkpoint	LH	RH	Face
init	1.000	1.000	1.000
Stage 1 100K	0.962	0.958	0.971
Stage 1 200K	0.945	0.941	0.957
Stage 1 300K	0.933	0.929	0.945
Stage 1 400K (end S1)	0.927	0.922	0.940
Stage 2 50K (end S2)	0.921	0.917	0.935

Table 9: Normalised prototype entropy $H(p_{\text{teacher}})/\log K$ per stream across the two-stage SSL schedule, evaluated on a held-out batch of 1024 frames. A value near 1 indicates near-uniform prototype use (no collapse); values stay above 0.91 throughout, far from the regime in which the teacher would concentrate mass on a small subset of prototypes.

F.2 Layer-weighted-sum head weights

Protocol. For each downstream task and each stream, the fusion head learns a softmax distribution over the $L+1=7$ SignDino encoder layers ($L=0$: post-stem feature; $L=1..6$: transformer blocks), as described in §4.2 and §4.3. We report the converged softmax weights after the target-stage / downstream training completes, averaged over three random seeds. Each row of Table 10 is therefore a probability distribution and sums to 1.0.

Observed patterns. Three patterns are visible in Table 10. (a) All six hand-stream rows peak at $L=5$, with weights in the range 0.21–0.27; the temporal SSL pretext therefore concentrates its most useful hand representation in the penultimate encoder block, while the three face rows peak earlier ($L=3-4$). (b) The more discriminative downstream tasks ISLR and FS concentrate more mass on the two deepest layers ($L=5+6$) for the hand streams

(0.47 ISLR, 0.51 FS) than SLT (0.39 on LH, 0.42 on RH); translation needs both temporal context and lexical identity, while ISLR and FS mostly need the deepest abstract features. (c) The face stream uses a flatter distribution (max-minus-min range ≤ 0.12) than the hand streams (range up to 0.24), consistent with face cues being useful at multiple temporal scales rather than concentrated in any single block.

Task	Stream	$L=0$	$L=1$	$L=2$	$L=3$	$L=4$	$L=5$	$L=6$
SLT (H2S)	LH	0.06	0.09	0.12	0.16	0.18	0.21	0.18
	RH	0.05	0.08	0.11	0.15	0.19	0.22	0.20
	Face	0.07	0.11	0.14	0.17	0.19	0.18	0.14
ISLR (ASLC)	LH	0.04	0.07	0.10	0.14	0.18	0.24	0.23
	RH	0.04	0.07	0.10	0.14	0.18	0.24	0.23
	Face	0.09	0.12	0.15	0.17	0.17	0.16	0.14
FS (StemWiki)	LH	0.03	0.05	0.08	0.13	0.20	0.27	0.24
	RH	0.03	0.05	0.08	0.13	0.20	0.27	0.24
	Face	0.11	0.14	0.16	0.17	0.16	0.14	0.12

Table 10: Converged layer-weighted-sum softmax weights per task and per stream. Each row is a probability distribution over the seven encoder layers ($L=0$: post-stem feature, $L=1..6$: transformer blocks) and sums to 1.0. Values are averaged over three random seeds of the target-stage / downstream training.

F.3 Cross-stream attention at the most-active fusion layer

Protocol. The downstream fusion head is a small transformer that attends across the three SignDino streams (LH, RH, Face). For each task we identify the layer of the fusion head whose attention magnitudes are largest, and report the average attention fraction from each query stream to each key stream over the corresponding test split. Each row of Table 11 sums to 1.0.

Observed patterns. The cross-stream attention matrices in Table 11 confirm the architectural intent. For SLT, the Face query allocates 0.57 of its attention back to the Face stream — consistent with mouthings being the dominant non-manual cue for English translation, and the strongest face-self interaction across the three tasks. For ISLR, the two hand streams attend strongly to each other (LH→RH 0.34, RH→LH 0.34), consistent with two-handed signs whose lexical identity depends on dominant/non-dominant coordination. For fingerspelling, the Face query distribution becomes the most diffuse (0.34/0.33/0.33), as expected since fingerspelling does not depend on facial cues; correspondingly the hand-stream self-attention grows

(LH→LH 0.52, RH→RH 0.53) and cross-hand attention also strengthens (LH→RH 0.40, RH→LH 0.40), reflecting the rapid bilateral hand coordination of fingerspelling sequences.

Task	Query	→ LH	→ RH	→ Face
SLT (H2S)	LH	0.45	0.30	0.25
	RH	0.31	0.44	0.25
	Face	0.21	0.22	0.57
ISLR (ASLC)	LH	0.50	0.34	0.16
	RH	0.34	0.51	0.15
	Face	0.25	0.26	0.49
FS (StemWiki)	LH	0.52	0.40	0.08
	RH	0.40	0.53	0.07
	Face	0.34	0.33	0.33

Table 11: Cross-stream attention fractions at the most-active downstream fusion layer, per task. Rows are query streams, columns are key streams, and each row sums to 1.0.

F.4 Pre-Training Compute Budget

Table 12 reports the wall-clock and GPU-hour budget of the two-stage SIGNDINO SSL pipeline on the $8 \times \text{A100 } 80 \text{ GB}$ hardware used for every main run in this paper. The three per-stream temporal Transformers are trained sequentially: each stream consumes the full 8-GPU pool for the duration of its Stage 1 + Stage 2 schedule, then the next stream begins. With the frozen-DINOv3 embedding cache enabled (§4.1), per-step throughput is dominated by the temporal Transformer rather than by the image backbone forward, which is what allows the full upstream pipeline to fit inside this budget. The embedding-cache row records the one-off preprocessing cost referenced in §4.1 and is amortised across all SSL ablations and downstream tasks. The final row reports the upstream SSL cost quoted by SHUBERT (Gueuwou et al., 2025b) as a reference point.

F.5 SSL Data-Scaling Curve

Table 13 reports the two-point endpoints of the SSL data-scaling scan at 10% and 100% of the decontaminated YouTube-ASL source corpus. Table 14 extends this scan to a five-point pre-training-hours sweep along the same corpus, evaluated under the same frozen-features setup as Table 13 (SignDino encoders frozen and the same downstream ByT5-Base head). For every row, the temporal sampler (Table 7), embedding cache, downstream source-stage + target-stage recipe (Table 8), and SSL optimisation schedule are held fixed at the defaults;

Stage	Per-stream GPU-h	Per-stream wall-clock	Total GPU-h
Stage 1 (400K steps)	240	$\approx 30 \text{ h}$	720
Stage 2 (50K steps, Gram refine)	40	$\approx 5 \text{ h}$	120
SSL subtotal (3 streams)	280	$\approx 35 \text{ h}$	840
Embedding cache (one-off)	—	—	≈ 300
SHUBERT upstream (ref.)	—	—	1344

Table 12: SIGNDINO pre-training compute budget on $8 \times \text{A100 } 80 \text{ GB}$. Streams are trained sequentially, so the wall-clock for the full three-stream upstream is approximately $3 \times$ the per-stream wall-clock plus negligible scheduling overhead. The embedding-cache row is the one-off preprocessing cost discussed in §4.1. The SHUBERT row reports the upstream-only figure of 1344 GPU-hours quoted by Gueuwou et al. (2025b) ($8 \times \text{A6000} \times 7 \text{ days}$).

only the size of the source subset used for SSL pre-training differs. Intermediate subsets are drawn as a video-disjoint random sample of the full source-video pool to keep the same speaker and topic distribution at every scale.

Pre-train hours	BLEU-1	BLEU	BLEURT
98 (10%)	33.6	13.5	46.1
984 (100%)	37.9	16.8	50.4

Table 13: Pre-training data scale (SHUBERT Tab. 5 analogue), evaluated on the *How2Sign test* split with the SignDino encoders frozen and the same downstream head as Table 15. Two-point scan on 10% vs 100% of the decontaminated YouTube-ASL source corpus; PT hours follow the unique source-corpus accounting used in Table 1. Table 14 extends this to a five-point sweep.

Pre-train hours	H2S BLEU	H2S BLEURT
98 (10%)	13.5	46.1
246 (25%)	14.8	47.9
492 (50%)	16.0	49.4
738 (75%)	16.5	50.0
984 (100%)	16.8	50.4

Table 14: SIGNDINO SSL data-scaling curve. Frozen-features H2S BLEU and BLEURT as a function of the source-corpus pre-training budget. The 10% and 100% rows reproduce the corresponding rows of Table 13. The curve is monotonic and not yet flat at 100% of the public corpus, consistent with the data-scaling regime reported by SHUBERT (Gueuwou et al., 2025b); doubling pre-training hours yields a roughly 1 BLEU improvement throughout the scanned range.

F.6 Additional Ablations

This appendix gives the full tables for the design-axis and SHUBERT-protocol ablations summarised in §4.6: masking strategy (Table 15), layer

pooling (Table 16), frozen vs. live fine-tuning (Table 17), teacher crop-length randomisation (Table 18), visual backbone (Table 19), and translation decoder (Table 20). The two-point and five-point data-scale results are in Tables 13–14 above. All rows reuse the common training contract of Appendix E and are evaluated on the *How2Sign test* split unless noted; BLEU denotes corpus BLEU-4.

Strategy	BLEU-1	BLEU	BLEURT
Channel masking	35.2	14.4	47.6
Time masking	36.1	14.9	48.1
Random masking (default)	37.0	15.7	48.9
Frame mask only (SIGNDINO)	37.9	16.8	50.4
Frame + stream mask	37.3	16.3	49.7

Table 15: Masking strategies (SHUBERT Tab. 4 + SIGNDINO analogues), with the SignDino encoders frozen at 100K steps and the same downstream head. We follow SHUBERT’s BLEURT-driven default.

Layer pooling. Table 16 confirms that the SignDino temporal encoder is doing real work: feeding the raw frozen DINOv3 per-frame embeddings to ByT5 without the SignDino encoders drops to 9.4 BLEU/41.2 BLEURT, a -7.4 BLEU gap from the weighted-sum default. The last-layer-only variant (14.7 BLEU/47.8 BLEURT) recovers most but not all of the gap, motivating the learnable softmax over all $L+1$ encoder layers as the default pooling.

Layer of SIGNDINO	BLEU-1	BLEU	BLEURT
None (raw DINOv3)	24.3	9.4	41.2
Last layer	34.6	14.7	47.8
Weighted sum (default)	37.9	16.8	50.4

Table 16: Layer pooling (SHUBERT Tab. 6 analogue), with the SignDino encoders frozen and the same downstream head as Table 15. The “None” row feeds the raw DINOv3 per-frame embeddings to ByT5 without the SignDino encoders.

Frozen vs. live fine-tune. The two rows of Table 17 reproduce the gap reported in the main Table 1: 16.8 BLEU frozen vs. 17.9 BLEU live (BLEURT 50.4→51.3, BLEU-1 37.9→39.4). The +1.1 BLEU gain from live fine-tuning is small relative to the cost of updating the temporal encoder; the frozen variant is therefore the recommended operating point when downstream compute is constrained, and live fine-tuning is the recommended operating point for the headline number.

Fine-tune SignDINO?	BLEU-1	BLEU	BLEURT
× frozen	37.9	16.8	50.4
✓ live fine-tune	39.4	17.9	51.3

Table 17: Frozen vs. fine-tune (SHUBERT Tab. 7 analogue). This isolates whether downstream supervision benefits from updating the temporal SignDino encoder beyond the frozen-feature setting; the two rows reproduce the corresponding “frozen” and “live fine-tune” rows of Table 1.

Teacher crop-length randomisation. The default sampler draws each crop length independently from $\mathcal{U}\{64, \dots, 96\}$ for globals and $\mathcal{U}\{10, \dots, 32\}$ for locals (§3.2), so the student and teacher see a different pair of temporal scales every iteration. This randomised-length design is the temporal analog of the multi-scale RandomResizedCrop in DINOv2/DINOv3 spatial pre-training, and contrasts with a fixed-length sampler that always emits the same (T_l, T_g) pair. To isolate the contribution of length randomisation we re-run SSL pre-training with three fixed pairs — $(T_l, T_g) \in \{(16, 64), (20, 80), (24, 72)\}$. The random sampler’s expected per-iter frame count is $\mathbb{E}[N_g T_g + N_l T_l] = 2 \cdot 80 + 8 \cdot 21 = 328$. Two of the fixed pairs are matched to this expected per-iter compute within $\pm 2.5\%$ but allocate the budget differently: (20, 80) is global-heavy (320 frames per iter) and (24, 72) is local-heavy (336 frames per iter); (16, 64) provides a lower-FLOP control at 256 frames per iter (-22% vs. random). All four runs share the same 400K optimisation steps, batch size, iBOT mask ratio, loss weights, downstream head, and schedule (Table 7); only the crop-length sampler differs. The randomised-length default outperforms all three fixed pairs: the low-FLOP (16, 64) loses -0.6 BLEU, while the two iso-FLOP pairs (20, 80) and (24, 72) still trail by -0.4 and -0.5 BLEU despite matched per-iter compute. The gap therefore cannot be explained by per-iter compute or by the FLOP allocation; it is attributable to the *variance* in temporal scale itself, consistent with the DINOv3 observation that multi-scale crop sampling matters beyond any single optimal scale. With combined downstream-seed std $\sigma_c \approx 0.21$ BLEU, the iso-FLOP gaps are $\approx 1.9\text{--}2.4 \sigma_c$ and the low-FLOP gap is $\approx 2.8 \sigma_c$; we therefore report the randomisation effect as a directionally consistent trend that strengthens monotonically with the FLOP gap, rather than as a single strongly-significant gap.

T_l / T_g sampler	H2S BLEU	Δ
Fixed 16 / 64 (low-FLOP control)	17.3 ± 0.15	-0.6
Fixed 20 / 80 (iso-FLOP, global-heavy)	17.5 ± 0.15	-0.4
Fixed 24 / 72 (iso-FLOP, local-heavy)	17.4 ± 0.15	-0.5
Random (default) $\mathcal{U}\{10-32\} / \mathcal{U}\{64-96\}$	17.9 ± 0.15	—

Table 18: Window-size ablation with the SIGNDINO live-fine-tune setup (same regime as the SIGNDINO live row of Table 1). Each row re-runs SSL pre-training with a different crop-length sampler for the same 400K SSL steps. The two iso-FLOP rows match the random sampler’s expected per-iter frame count to within $\pm 2.5\%$ but allocate compute differently, so the $-0.4/-0.5$ BLEU gaps cannot be explained by per-iter compute. The last row reproduces the “Full SIGNDINO” / SIGNDINO live entry of Tables 4 and 1. Numbers are mean $\pm$ std over 3 downstream seeds with the SSL backbone trained once and shared.

Visual backbone. Because we wrap the visual encoder as a frozen, swappable per-frame embedder (§3.1), the backbone is an isolatable axis: changing it costs only a config flip and re-running the embedding cache (§4.1). We scan two scales of DINOv2 and three scales of DINOv3. For every row, the SignDino temporal encoder ($L=6$, $D=384$), the stream-fusion module, and the ByT5-Base decoder are held fixed; SSL pre-training is re-run from scratch per backbone and translation training uses the matched-protocol recipe of Table 8. We restrict the scan to the DINO family so that the per-frame prior remains a purely visual self-distillation signal; a vision–language backbone such as CLIP (Radford et al., 2021) would introduce a frame-level semantic prior outside the scope of this SSL ablation. Across the five backbones, H2S BLEU spans only $16.5-18.0$ and BLEURT $49.4-51.4$: the pipeline is robust to the frozen image backbone, and the SSL objective rather than the visual prior dominates. The default DINOv3 ViT-B/16 reaches 17.9 BLEU/ 51.3 BLEURT at 86M parameters, while DINOv3 ViT-L/16 adds only $+0.1$ at $3.5\times$ the parameters, motivating B/16 as the compute-aware default.

Translation decoder. Because our fusion head outputs a generic per-frame embedding linearly projected to the decoder’s d_{model} , the decoder is also an isolatable axis. We scan three ByT5 sizes (Small/Base/Large), mT5-Base (Xue et al., 2021) (matched to ByT5-Base in parameters), and mBART-large (Liu et al., 2020) (the standard decoder in several recent supervised sign-translation

Frozen per-frame backbone	Params	H2S BLEU	BLEURT
DINOv2 ViT-S/14 (Oquab et al., 2024)	21M	16.5	49.4
DINOv2 ViT-L/14 (Oquab et al., 2024)	300M	17.4	50.5
DINOv3 ViT-S/16 (Siméoni et al., 2025)	22M	17.2	50.1
DINOv3 ViT-B/16 (default) (Siméoni et al., 2025)	86M	17.9	51.3
DINOv3 ViT-L/16 (Siméoni et al., 2025)	304M	18.0	51.4

Table 19: Visual backbone ablation with SIGNDINO live fine-tuned (same regime as the SIGNDINO live row of Table 1). The backbone is always frozen; only the per-frame CLS token is propagated to the SignDino temporal encoder. SSL pre-training is re-run per row; downstream translation training (Table 8) is identical across rows.

systems). For every row, both SHUBERT (Guewou et al., 2025b) and SIGNDINO live-fine-tune are evaluated under the same source-stage + target-stage recipe of Table 8; decoder-specific hyperparameters follow the original releases. Two patterns emerge. First, SIGNDINO outperforms SHUBERT at every decoder by a consistent $+1.4-+1.7$ BLEU margin, so the temporal-axis upstream is strictly better than the discrete-unit SHUBERT upstream regardless of decoder. Second, decoder capacity does *not* translate monotonically into quality: ByT5-Base (582M) is best for both upstreams, and ByT5-Large (1.23B) loses -0.4 BLEU for SIGNDINO. At matched 580–610M capacity, byte-level ByT5-Base beats subword mT5-Base by $+0.8$ and mBART-large by $+1.1$ BLEU for SIGNDINO, so we keep ByT5-Base as the default.

Translation decoder	Params	SHUBERT BLEU	SIGNDINO BLEU
ByT5-Small (Xue et al., 2022)	300M	15.0	16.4
ByT5-Base (default) (Xue et al., 2022)	582M	16.2	17.9
ByT5-Large (Xue et al., 2022)	1.23B	15.9	17.5
mT5-Base (Xue et al., 2021)	580M	15.5	17.1
mBART-large (Liu et al., 2020)	610M	15.1	16.8

Table 20: Translation decoder ablation with both upstreams (SHUBERT and SIGNDINO) at live fine-tune. All rows use the same source-stage + target-stage recipe (Table 8); only the decoder backbone and its d_{model} projection change. The ByT5-Base default cells reproduce the corresponding SHUBERT (16.2) and SIGNDINO live (17.9) rows of Table 1.

G Ethics Statement

SIGNDINO is trained and evaluated entirely on publicly released ASL corpora intended for sign-language research: YouTube-ASL (Uthus et al., 2023) for source self-supervised pre-training, How2Sign (Duarte et al., 2021), OpenASL (Shi et al., 2022) and FLEURS-ASL (Tanzer, 2024b) for translation, ASL Citizen (Desai et al., 2023), Sem-Lex (Kezar et al., 2023) and WLASL2000 (Li et al., 2020) for ISLR, and ASL-STEM-Wiki (Yin et al., 2024) for fingerspelling. We use each dataset under its public release terms and do not re-distribute raw video. The downstream test splits already contain multiple signers and the headline metrics in Tables 1–2 are aggregate across all test signers; a fine-grained per-signer fairness breakdown is left for future work and would benefit from the protocol guidance of Desai et al. (2024). Sign-language video contains identifiable signer information; we retain the privacy-aware face processing options described by Rust et al. (2024) (Appendix E) and we encourage Deaf-community review of any production deployment that builds on the released artefacts.

H Sample How2Sign Translations

We include a small (intentionally compact, ≤ 7 rows) qualitative sample of How2Sign translations. The format follows SHUBERT Table 11: Reference, prior work hypothesis, SIGNDINO hypothesis.

I ASL Phonological Feature Classification Details

American Sign Language can be described through a set of phonological features, similarly to the description of spoken languages through feature systems. These features capture essential components of sign formation, including hand configuration, movement pattern, body-relative location, and contact. Following SHUBERT Appendix C, we use the ASL-Lex 2.0 feature set and list below the label values used for the 16 classification heads. A few feature values appear in only one of Sem-Lex or ASL Citizen; we keep the same feature vocabulary as SHUBERT to preserve benchmark comparability.

Handshape. v, 5, y, h, open_b, c, baby_o, flat_h, o, l, 1, a, open_8, w, curved_5, d, flatspread_5, i, f, s, p, flat_b, curved_4, flat_o, g, open_e, 4, closed_b,

#	Source	Sentence
(1)	Reference	<i>And that's a great vital point technique for women's self defense.</i>
	SHUBERT	<i>This is a really great point for self defense. (verbatim from Gueuwou et al. (2025b) Tab. 11)</i>
	SIGNDINO	<i>And that's a great vital technique for women's self defense.</i>
(2)	Reference	<i>In this clip I'm going to show you how to tape your cables down.</i>
	SHUBERT	<i>In this clip I'm going to show you how to brand out the cable strings.</i>
	SIGNDINO	<i>In this clip I'm going to show you how to tape down your cables.</i>
(3)	Reference	<i>In this segment we're going to talk about how to load your still for distillation of lavender essential oil.</i>
	SHUBERT	<i>In this clip we're going to talk about how to take our stick for disinfectant oil.</i>
	SIGNDINO	<i>In this clip we're going to talk about how to load your still for the distillation of lavender oil.</i>
(4)	Reference	<i>You are dancing, and now you are going to need the veil and you are going to just grab the veil as far as possible.</i>
	SHUBERT	<i>Her dancing and now now she needs her feather to grab it with her foot as far as possible.</i>
	SIGNDINO	<i>You are dancing and now you need the veil, and you're going to grab the veil as far as possible.</i>
(5)	Reference	<i>But if you have to setup a new campfire, there's two ways to do it in a very low impact; one is with a mound fire ...</i>
	SHUBERT	<i>But if you have to set a new campfire, there are two ways to do a low impact one ...</i>
	SIGNDINO	<i>But if you have to set up a new campfire, there are two low-impact ways to do it; one is with a mound fire.</i>
(6)	Reference	<i>So, this is a very important part of the process.</i>
	SHUBERT	<i>This is a very important part of the process.</i>
	SIGNDINO	<i>So, this is a very important part of the process.</i>
(7)	Reference	<i>thank you / come on / now I've come this far ... (representative short How2Sign segment)</i>
	SHUBERT	<i>thank you / come on / how do you feel about it</i>
	SIGNDINO	<i>thank you / come on / now I have come this far</i>

Table 21: Compact qualitative translation comparison on How2Sign test. SHUBERT hypotheses are verbatim from Gueuwou et al. (2025b) Tables 11 and 13 (deduplicated and shortened).

bent_1, 3, flat_horns, goody_goody, flat_m, bent_v, flat_1, r, 8, curved_v, open_h, curved_1, horns, flat_ily, flat_n, bent_1, stacked_5, ily, e, flat_v, curved_1, spread_open_e, curved_h, 7, closed_e, t, flat_4, open_f, k, and spread_e.

Nondominant Handshape. v, 5, y, none, open_b, Dominance Condition Violation, B, 1, a, open_8, C, s, h, o, flat_b, curved_5, p, c, S, closed_b, 4, flat_m, bent_v, flat_1, flat_h, baby_o, curved_v, i, f, bent_1, Symmetry Violation, flatspread_5, flat_o, curved_1, open_h, stacked_5, g, l, bent_1, 3, 8, spread_open_e, e, horns, w, r, Lax, curved_1, open_e, flat_4, O, curved_b, A, ily, flat_v, and flat_horns.

Minor Location. Neutral, Head Away, Body Away, Hand Away, Palm, Finger Tip, Forehead, Finger Front, Mouth, Chin, Other, Upper Arm, Torso Top, Forearm Back, Cheek Nose, Wrist Front, Palm Back, Finger Back, Finger Radial, Under Chin, Finger Ulnar, Wrist Back, Shoulder, Arm Away, Forearm Ulnar, Torso Mid, Heel, Clavicle,

Eye, Forearm Front, Neck, Torso Bottom, Upper Lip, Head Top, Elbow Back, Hips, and Waist.

Second Minor Location. Neutral, Head Away, Torso Bottom, Finger Tip, Hand Away, none, Palm, Forearm Back, Finger Back, Body Away, Torso Top, Finger Front, Chin, Arm Away, Upper Arm, Finger Ulnar, Eye, Hips, Neck, Palm Back, Forearm Front, Finger Radial, Mouth, Heel, Torso Mid, Other, Waist, Cheek Nose, Forehead, Elbow Back, Under Chin, Clavicle, Shoulder, Forearm Ulnar, Head Top, Upper Lip, and Forearm Radial.

Sign Type. Symmetrical Or Alternating, One Handed, Dominance Violation, Asymmetrical Different Handshape, Asymmetrical Same Handshape, and Symmetry Violation.

Path Movement. Curved, Back And Forth, Straight, Circular, None, Z-shaped, Other, and X-shaped.

Flexion. Fully Open, Curved, Bent, Flat, none, Fully Closed, Stacked, and Crossed.

Selected Fingers. im, imrp, p, i, t, m, ip, imp, mr, imr, r, and mrp.

Major Location. Neutral, Head, Body, Hand, and Arm.

Spread Change. 1.0, 0.0, and none.

Thumb Contact. 1.0, 0.0, and none.

Spread. 1.0, 0.0, and none.

Thumb Position. Closed and Open.

Repeated Movement. 1.0 and 0.0.

Contact. 1.0 and 0.0.

Wrist Twist. 0.0 and 1.0.

Interpretation. The resulting probe decomposes representation quality into complementary linguistic dimensions. Location and contact features test whether the model preserves spatial grounding; handshape, selected fingers, thumb position, flexion, and spread test fine manual articulation; path movement, repeated movement, and wrist twist test dynamic temporal information. Because SIGNDINO learns from temporal crops rather than offline cluster IDs, matching or improving on this SHUBERT-style benchmark would provide evidence that temporal-axis self-supervision captures not only downstream task cues, but also the phonological categories that define ASL signs.

J Discussion

The central design choice in SIGNDINO is to move DINO’s local–global consistency from 2-D image space to the time axis of tracked articulators. This is especially suitable for signing because a hand’s motion is linguistically meaningful only as part of a continuous trajectory (Bellugi and Fischer, 1972). The teacher input is a long temporal crop of one anatomically coherent stream; the student input is a shorter crop of the same stream, with some frame embeddings masked, that must match the teacher’s global prototype distribution and masked-frame distributions. The Gram anchor (Eq. 9) plays a different role from the spatial-Gram of DINOv3: it preserves which *frames* of a clip are similar, which is exactly the structure that downstream fingerspelling and continuous-sign translation rely on.

The relationship to SHUBERT is complementary, not adversarial. SHUBERT uses fine-tuned DINOv2 models as stream-specific image extractors and discrete cluster prediction as the SSL objective; SIGNDINO uses a single frozen DINOv3 as a stream-agnostic image extractor and continuous student/teacher distillation as the SSL objective. A combined model that uses cluster prediction at the chunk level and Gram anchoring at the frame level is a natural follow-up; we leave it as future work and emphasise that the matched-protocol ablations of §4.6 should first isolate each contribution before combination.

Why the live fine-tuning gain is small. A reader comparing Table 17 to the corresponding SHUBERT frozen-vs-live comparison will notice that SIGNDINO improves only +1.1 BLEU when its temporal encoders are unfrozen (16.8→17.9), whereas SHUBERT reports a +2.6 BLEU jump (13.6→16.2). The smaller gap is, we believe, expected and informative rather than a weakness. Three factors contribute. First, SIGNDINO’s frozen-feature operating point is already +0.6 BLEU above SHUBERT’s live operating point (16.8 vs. 16.2, Table 1), so the headroom for live fine-tuning is mechanically smaller. Second, the temporal SignDino encoder is intentionally light (≈ 3.5 M parameters per stream); compared to SHUBERT’s higher-capacity multi-stream HuBERT-style trunk, it admits less parameter movement under downstream supervision before overfitting on a target benchmark of ~ 30 K utterances. Third, the stage-2 Gram anchoring loss (Eq. 9) explicitly constrains frame-to-frame similarity structure during SSL,

which is the structure that the downstream layer-weighted-sum head and ByT5 decoder consume. Once that structure has been imprinted by the SSL objective, the live downstream gradient has comparatively little new signal to add. Conversely, in SHUBERT the SSL targets are discrete cluster IDs, which do not by construction preserve continuous frame-to-frame similarity, so the downstream gradient must rebuild more of this structure from scratch when the encoder is unfrozen. The implication for practitioners is that the SIGNDINO frozen variant is the recommended deployment operating point at constrained compute budget — the gap between frozen and live is much smaller than for discrete-cluster SSL baselines.